Wage Inequality in Turkey

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Contents

1 Introduction and Definition **2**

2 Data and Methodology **5**

3 Results **9**

3.1 Findings and Comments 9

3.2 Comparison with the World 18

4 Review of the Literature and Conclusions **19**

5 References **25**

1 Introduction and Definition

The purpose of this study is to understand wage inequality in Turkey. It aims to explain how wages are distributed among workers and why some people earn much less than others. By using recent data, the study looks at the size of wage gaps, who is most affected, and what causes these differences. It also compares Turkey to other countries to show how serious the problem is. The goal is to help people and policymakers see why wage inequality matters and what can be done to reduce it. In short, this study helps us learn more about fairness, opportunity, and economic justice in the Turkish labor market.

Wage inequality means that people earn different amounts of money for their work. Some workers earn a lot, while others earn very little. For example, a factory worker and a manager in the same company can have very different wages. This is a big problem in many countries like Turkey where wages are the main income source for most people. It affects how people live, how the economy grows, and how fair the society is.

Wage inequality is different from income or wealth inequality. Income inequality includes all types of income, such as wages, profits, and government support. Wealth inequality is about who owns houses, land, and savings. But wage inequality only focuses on what people earn from working. This makes it very important to study, especially when we want to understand problems in the labor market. Table 1 summarizes the main differences among wage, income, and wealth inequality.

Table 1: Comparison of Wage, Income, and Wealth Inequality

	What it Measures	**Main Components**
Wage Inequality	Differences in salaries/wages among workers	Hourly/monthly earnings from labor
Income Inequality	Differences in total income across individuals	Wages, self-employment income, capital income, transfers
Wealth Inequality	Differences in assets and net worth	Property, savings, stocks, inheritance

Note: Wage inequality is a part of income inequality, whereas wealth inequality is usually more extreme and persistent over generations.

There are many ways to measure wage inequality. One is the Gini coefficient. A Gini score of 0 means everyone earns the same. A score of 1 means one person earns everything. Other tools include comparing the top and bottom wages (like the P90/P10 ratio), or calculating the variance in wages. These numbers help us understand how big the wage gap is.

As shown in Figure 1, nearly 40% of workers in Turkey earn less than 1.05 times the minimum wage (Cunedioğlu, 2023). This is the highest rate among the countries shown. In places like Slovenia and Bulgaria, this share is only around 13% to 15%. In most EU countries, it is even lower—under 10%. This means that a large number of Turkish workers are stuck at the bottom of the wage ladder. Their wages are very low, and they have few chances to move up. This is called wage compression. It means that wages are not fairly spread across workers. It also shows that the labor market in Turkey offers limited upward mobility. So, wage inequality in Turkey is not only about the rich earning more. It is also about millions of workers being trapped in low-paying jobs.

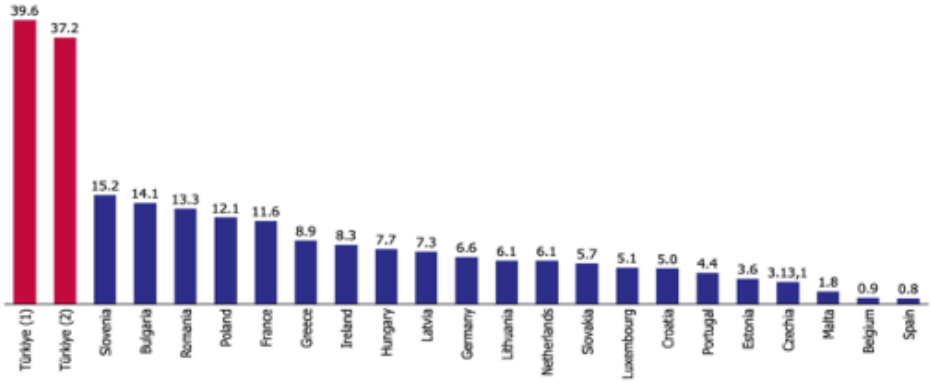


Figure 1: Share of wage earners and casual employees earning less than 1.05 times the minimum wage in total wage earners and casual employees (%, 2018)

Source: Cunedioğlu (2023), based on Eurostat data.

Many factors cause wage inequality. Education is a big one. People with more education often earn more. Skills and work experience also matter. Technology has made this worse as new machines and computers increase the need for skilled workers, but leave others behind. Another problem is Turkey's labor market because there are many informal jobs, and unions are weak. Globalization also plays a role. Companies move jobs to cheaper countries, putting pressure on local wages. Table 2 outlines these primary

drivers and their impacts, which interact and reinforce inequality.

Table 2: Main Factors Causing Wage Inequality

Factor	**Impact**
Human Capital	Education, skills, and experience gaps
Technological Change	Higher demand for skilled workers
Labor Market Structure	Informality and weak unions
Globalization	Outsourcing pressures

Wage inequality does not affect everyone equally. Some groups suffer more. Women, for example, still earn less than men for similar jobs. However, in our data, the unadjusted gender wage premium appears to be close to one. This means that, on average, women and men who are employed as positive wage earners earn similar wages in Turkey according to recent statistics, though sample selection plays a critical role. Young people struggle to find good jobs. People in eastern and southeastern Turkey have fewer opportunities. These groups face long-term structural disadvantages, making it hard to escape poverty.

Wage inequality causes many problems. First, it slows down the economy. People with low wages cannot spend much, which reduces demand. Second, it makes it harder for people to move up in life. If you are born poor, you stay poor. Third, it creates anger and tension in society. People feel things are unfair. This weakens trust in the government and democracy, and creates intergenerational cycles of disadvantage.

Many economists have studied inequality. Thomas Piketty (2014) shows that capital and wealth grow faster than wages, so rich people get richer. Anthony Atkinson argues that we need better education and fair taxation policies. Raj Chetty et al. (2017) show that inequality makes it hard for people to improve their lives and lowers intergenerational mobility. In the Turkish context, Orhan Torul highlights Turkey's persistent link between education and informality in shaping wage disparities (Tamkoç and Torul, 2020; Torul and Öztunalı, 2018).

There are also some wrong ideas about inequality. Some say, "If you work hard, you will earn more." But this is not always true. Opportunities are not equal for everyone. Others believe that inequality makes people work harder. But in many cases, it does the opposite. When people feel they cannot win, they stop trying. Some think inequality is

natural in a market economy. But policies and laws shape the market. Governments can reduce or increase inequality with their choices. Finally, inequality does not only affect the poor; it harms the whole society.

It is also important to know that wage inequality does not tell us everything. A country can be rich and still have high inequality, like the United States. On the other hand, low wage inequality does not automatically mean economic fairness or prosperity. Many factors affect inequality, such as automation, policy changes, and migration, so it is not a standalone measure of economic health. Wage inequality is just one part of the story, but it is an important one.

2 Data and Methodology

To construct the ground on which the analyses depend, what the data are, how to classify and make them usable for the estimation, and how to deal with them will be examined. For the analysis, two different official datasets will be used: the Household Budget Survey (HBS) and the Survey of Income and Living Conditions (SILC), which are valid data for the methodology that will be used. The same methodological framework with Review of Economic Dynamics (RED) guidelines for inequality studies will be applied. Moreover, the to-be-calculated inequality measurements, inequality premiums, and annual earning estimations also comply with RED guidelines.

To be able to analyze the evolution of wage inequality in Turkey since 2002, HBS and SILC are used extensively. Both datasets come from the Turkish Statistical Institute (TurkStat), which is the official statistical institute. HBS and SILC are annual surveys that collect data on income, consumption, and earnings from households and are constructed to be representative of Turkey. For the estimation for 2022, HBS in 2022 and SILC in 2023 are used (the publication year in SILC reflects the previous year's income data). HBS data includes disposable income and consumption subcomponents at the household level and labor supply and income sources at the individual level, such as labor earnings, asset income, self-employment, and government/private transfers. The focus is on labor earnings and self-employment income. HBS surveys 39,189 persons in 2022 and is nationally representative. SILC has been conducted since 2006 and covers detailed income sources for all household members. Labor earnings and self-employment income are focused on here too. SILC surveys 60,223 persons in 2023 (for 2022 refer-

ence period) and is nationally representative. Both HBS and SILC comply with RED guidelines for inequality studies (Tamkoç and Torul, 2020).

For the estimation, all monetary variables are converted to real terms using CPI (2002 base). Because wages are concerned, working-age individuals (25–60 years) are included, while individuals who do not report hours worked and individuals whose annual earnings are below 2,439 TL are excluded. This threshold is used because 2,439 TL in 2022 is approximately equal to 50% of Turkey's monthly minimum wage at the time, allowing the analysis to focus on the wage inequality of individuals who work regularly.

To be compatible with RED guidelines, annual earnings are calculated as follows:

$$a_{e,i,t} = nw_{i,t} + rw_{i,t} + p_{i,t} + b_{i,t} + \alpha_{\text{TUR}}(nse_{i,t} + rse_{i,t} + ag_{i,t})$$

where $a_{e,i,t}$ denotes annual earnings, $nw_{i,t}$ and $rw_{i,t}$ denote annual cash and other real (in-kind) payments, $p_{i,t}$ and $b_{i,t}$ denote annual premiums and bonuses received from employers, α_{TUR} denotes the share of labor income in Turkey's national income ($\sim 42\%$),[1] $nse_{i,t}$ and $rse_{i,t}$ denote cash and other real incomes from self-employment, and $ag_{i,t}$ denotes net agricultural income. These components are summarized in Table 3.

Table 3: Components of Annual Earnings Calculation

Symbol	**Description**
$nw_{i,t}$	Annual cash payments
$rw_{i,t}$	Annual real (in-kind) payments
$p_{i,t}$	Annual premiums
$b_{i,t}$	Annual bonuses
α_{TUR}	Labor share of national income ($\sim 42\%$)
$nse_{i,t}$	Cash self-employment income
$rse_{i,t}$	Real self-employment income
$ag_{i,t}$	Net agricultural income

To make the estimation richer and to make comparisons between subgroups in the population, wage premiums are calculated. To that end, hourly wage is used. Hourly

[1] As of 2019, the labor share of national income in Turkey is calculated as $\approx 43\%$ with a decreasing trend. Source: `https://fred.stlouisfed.org/series/LABSHPTRA156NRUG`.

wage is calculated as annual earnings divided by annual hours worked. The gender wage premium measures the ratio of average male wages to average female wages. The education (college) premium represents the relative earnings of college-educated workers compared to high school graduates, reflecting both productivity gains from education and credential effects. The experience premium captures the wage differences associated with accumulated work experience by comparing older (ages 45–55) and younger (ages 25–35) male workers. Hourly wage and the wage premiums are calculated as follows:

Hourly Wage Calculation:

$$w_{i,t} = \frac{ae_{i,t}}{ah_{i,t}}$$

where $ae_{i,t}$ is annual earnings and $ah_{i,t}$ is annual hours worked.

Table 4: Definitions of Wage Premiums

Premium Type	**Definition**
Gender Premium	$\overline{w_{m,t}}/\overline{w_{f,t}}$ (Male / Female avg. wage)
College (Education) Premium	$\overline{w_{c,t}}/\overline{w_{h,t}}$ (College / HS grad males)
High School Premium	$\overline{w_{h,t}}/\overline{w_{s,t}}$ (HS / Secondary school males)
Experience Premium	$\overline{w_{o,t}}/\overline{w_{y,t}}$ (Age 45–55 / 25–35 males)

Note: $\overline{w_{x,t}}$ denotes the average hourly wage for group x in year t.

Gini coefficient, variance of residuals, percentile ratios (P90/P50, P50/P10), and variance of log wage are used to estimate wage inequality in the country. These differ in that each assigns different weights to different parts of the population. The variance of residuals measures wage inequality that persists after accounting for observable worker characteristics (e.g., education, age, gender, family composition) in a regression model. It captures disparities stemming from unobserved factors such as discrimination, unmeasured skills, labor market imperfections, or pure randomness. The Gini coefficient quantifies overall wage inequality across the entire distribution, ranging from 0 (perfect equality) to 1 (one person earns all income). It is derived from the Lorenz curve, which plots cumulative wage shares against the cumulative share of the population (Figure 2). Specifically, the Gini coefficient is equal to the ratio of the area between the line of absolute equality and the Lorenz curve (A) to the total area under the line of absolute equality

$(A + B)$, i.e., $G = \frac{A}{A+B}$ (or $2A$ when the unit square is normalized so that $A + B = 0.5$).

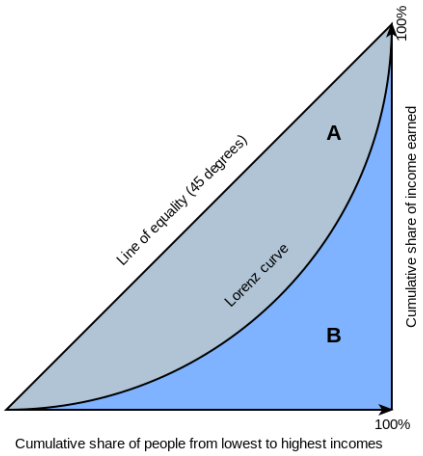


Figure 2: Lorenz curve and the line of absolute equality ($G = \frac{A}{A+B}$)

Percentile ratios dissect inequality by comparing specific points in the wage distribution: P90/P50 (top 10% vs. median) and P50/P10 (median vs. bottom 10%) isolate upper- and lower-tail inequality, respectively. Finally, the variance of log wages measures total proportional wage dispersion using log-transformed wages, which reduces the influence of extreme values and skewness. Variance of residuals and variance of log wage are calculated as follows:

- **Variance of Residuals:**

$$\log(y_{i,t}) = c_t + \beta_1 D^s_{i,t} + \beta_2 D^e_{i,t} + \beta_3 D^f_{i,t} + \beta_4 D^a_{i,t} + \beta_5 D^{ce}_{i,t} + \epsilon_{i,t}$$

  Then, we calculate the variance of $\hat{\epsilon}_{i,t}$, the residual error term.

- **Control Variables:**

Symbol	Description
D^s	Gender dummy
D^e	11 education dummies
D^f	Family composition dummies
D^a	7 age category dummies
D^{ce}	4 couple education dummies

- **Variance of Log Wage:** $\text{Var}(\log W) = \frac{1}{N}\sum_{i=1}^{N}(\log w_i - \overline{\log w})^2$

3 Results

3.1 Findings and Comments

In this section, we examine the general condition of wage inequality and the socioeconomic factors affecting wage inequality in Turkey. First of all, to make the analysis academically coherent with previous research, we followed the RED guidelines to assess the wage earners in both the SILC and HBS datasets. We included individuals between 25 and 60 years old. We excluded individuals who did not report how many hours they worked. Finally, we excluded individuals whose annual earnings are below 2,439 Turkish Liras (i.e., half of Turkey's monthly minimum wage in 2022; Ministry of Labor and Social Security, 2022).

To begin with, we constructed a descriptive statistics table to understand the main hourly wage metrics from both datasets (Table 5). The most notable factor is the difference in nominal mean and median wages between the two datasets, while 1% trimming is applied to HBS to remove extreme outliers. Moreover, while the mean wages are very close between gender groups, the maximum values belong to male earners in both datasets.

Table 5: Descriptive Statistics of Hourly Wages (2022)

Group	**Min**	**Max**	**Mean**	**Median**	**Std. Dev.**
SILC - Overall	1.3846	1079	42.7545	31.7307	44.1467
SILC - Male	1.3846	1079	43.2685	31.7692	46.4847
SILC - Female	1.4690	807	41.3580	31.1965	37.0245
HBS - Overall	5.8080	633	136.8628	115.9861	99.4841
HBS - Male	5.8500	633	137.4669	116.6666	98.3245
HBS - Female	5.8080	625	135.1767	112.5000	102.6521

Note: All values are in TL. A 1% threshold is removed from HBS to remove outliers.

We conducted a comprehensive analysis capturing the main metrics of wage inequality for both datasets. Specifically, we calculated the Gini coefficient, variance of log wages, variance of residuals, education, gender, and experience premiums, P90/P50, and P50/P10 values (Table 6), and compared our results with the updated 2022 estimations of Orhan Torul and Mehmet Nazım Tamkoç (Tamkoç and Torul, 2020) shown in Table

7. Our results closely align with their estimations, with only minor differences.

	Gini	Var(Log Wages)	Edu. Premium	Gender Premium	Exp. Premium	P90/P50	P50/P10	Var(Residuals)
SILC	0.4120	0.6304	1.6463	1.0461	1.0509	2.5454	2.8448	0.4605
HBS	0.4304	0.6983	1.5623	1.0050	0.9175	2.3535	2.4871	0.5058

Table 6: Our estimations for 2022

	Gini	Var(Log Wages)	Edu. Premium	Gender Premium	Exp. Premium	P90/P50	P50/P10	Var(Residuals)
SILC	0.4241	0.6580	1.5441	1.0265	1.0037	2.5013	2.7165	0.4898
HBS	0.4173	0.6480	1.5884	1.0095	0.9046	2.3413	2.8752	0.5902

Table 7: Torul's estimations for 2022

Figure 3 displays our estimated Lorenz curves for hourly wages in HBS (Gini = 0.43) and SILC (Gini = 0.412). In both surveys, the curvature away from the 45-degree line of equality illustrates substantial wage dispersion, where the bottom half of the workforce receives less than a quarter of total hourly wages.

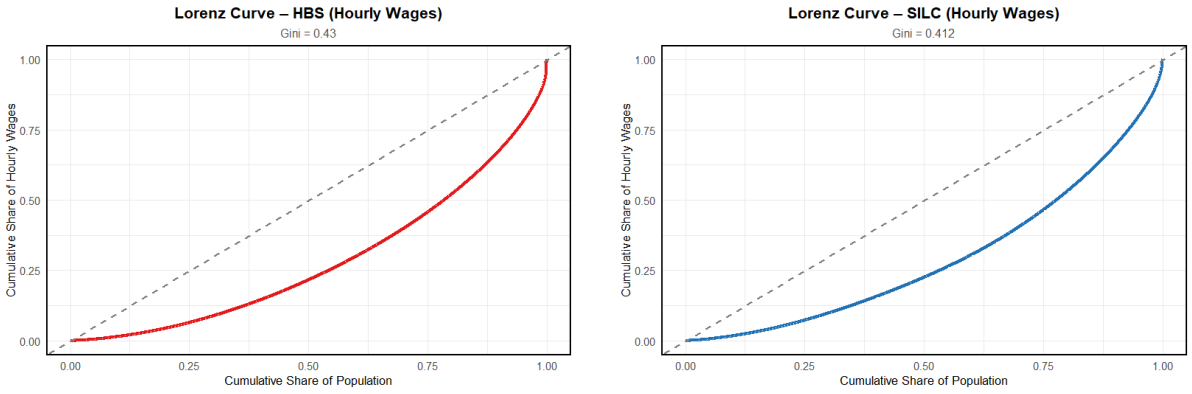


Figure 3: Lorenz Curves for HBS and SILC Hourly Wages

Furthermore, we deepened our analysis by looking at the regional average hourly wage and regional Gini coefficients using the SILC dataset. First, we look at the regional average hourly wages. As can be seen from Figure 4, the regions with the lowest average hourly wages are Central East Anatolia (TRB, 33 TL) and Southeast Anatolia (TRC, 35 TL), while Istanbul (TR1, 53 TL) and West Anatolia (TR5, 52 TL) have the highest average hourly wages. These results also correlate with the SEGE-2017 socio-economic development rankings (Acar et al., 2019), which show that the least developed regions in Turkey are Southeast and East Anatolia.

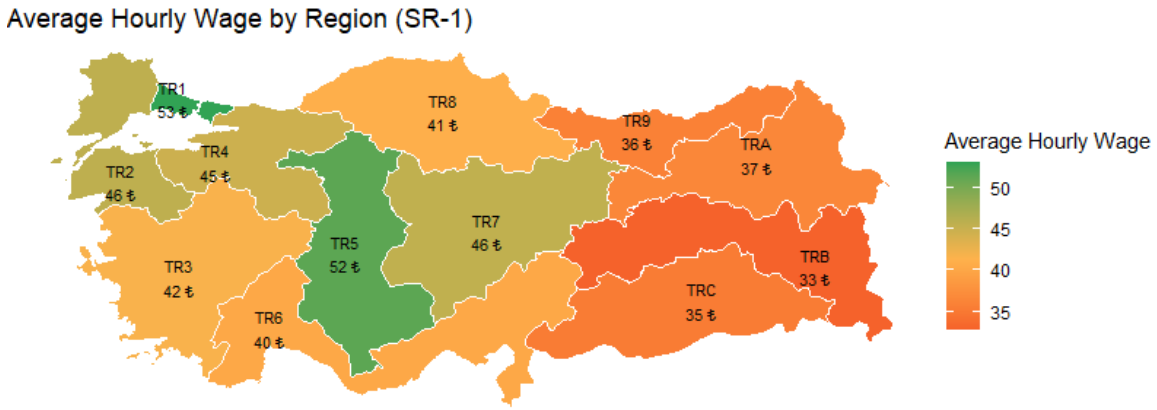


Figure 4: Regional Analysis of Average Hourly Wage (SR-1)

We also examined the regional Gini coefficients. Figure 5 shows wage inequality within the boundaries of SR-1 regions. Notably, the ranking of within-region inequality differs from the ranking of average : the most unequal regions are in the northern and northeastern parts of Anatolia, particularly the East Black Sea region (TR9, Gini = 0.469) and Northeast Anatolia (TRA, Gini = 0.459), whereas East Marmara (TR4, Gini = 0.344) has the lowest within-region Gini coefficient.

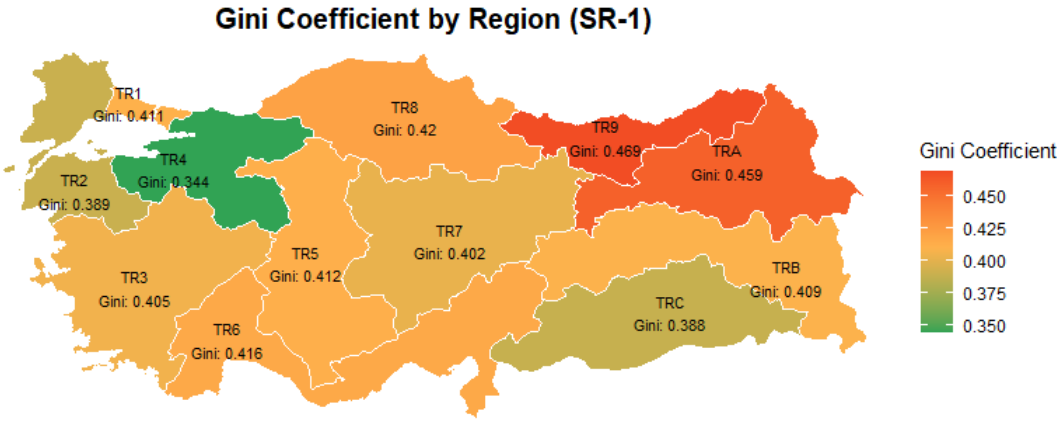


Figure 5: Regional Analysis of Gini Coefficient (SR-1)

Next, we investigated wage differences between the public and private sectors using the SILC dataset. As shown in Figure 6, public sector workers earn higher average hourly wages than private sector workers across every decile of the wage distribution. In the private sector, hourly wages grow slowly across the first eight deciles due to heavy clustering around the minimum wage, followed by a sharp spike in the 9th and 10th

deciles. By contrast, public sector wages increase more steadily across deciles, narrowing the gap with the private sector only at the top decile.

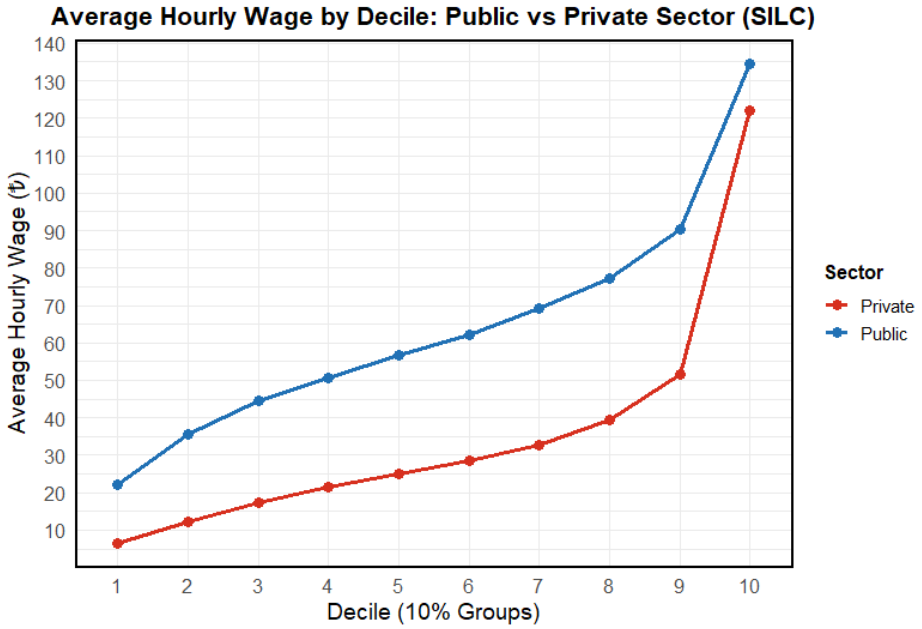


Figure 6: Average Hourly Wage by Decile: Public vs. Private Sector (SILC)

To evaluate the role of education in wage inequality, we calculated the Gini coefficients separately for university and non-university graduates in both datasets (Table 8) and plotted average hourly wages across deciles by education group (Figure 7). In both SILC and HBS, wage inequality is lower among university graduates (0.3398 in SILC and 0.3124 in HBS) than among non-university workers (0.3645 in SILC and 0.3523 in HBS). At the same time, Figure 7 demonstrates that university graduates earn substantially higher hourly wages at every decile, with the gap widening dramatically in the top deciles.

Table 8: Gini Coefficients by Education Level

Group	**Gini**
SILC - University	0.3398
SILC - Non-University	0.3645
HBS - University	0.3124
HBS - Non-University	0.3523

- For SILC, 4-year higher education and above are counted as University.
- For HBS, Higher Educational Institutions for 2-3 years are counted as University.

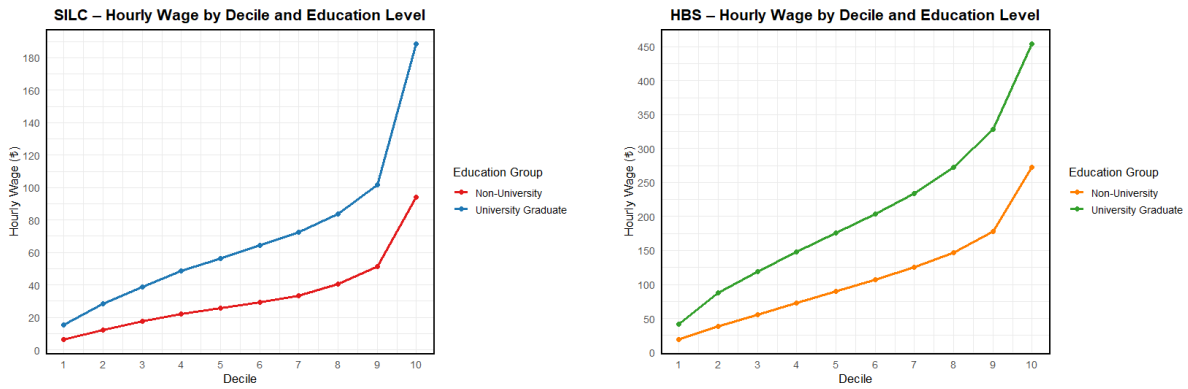


Figure 7: Hourly Wage by Decile and Education Level in SILC (left) and HBS (right)

To understand wage inequality across age groups, we also calculated the Gini coefficient for three age brackets: 25–35, 35–45, and 45–60 (Table 9), and plotted their hourly wage profiles by decile (Figure 8). While there are slight level differences in Gini coefficients between the two datasets, their pattern is identical: in both SILC and HBS, the Gini coefficient is lowest in the youngest group (25–35) and increases monotonically with age, reaching 0.4641 (SILC) and 0.4142 (HBS) in the 45–60 age group. As people first enter the labor market, their earnings are relatively similar due to limited experience; over time, as workers accumulate experience and specialize, earnings diverge and within-cohort wage inequality grows. Figure 8 also shows that the 45–60 group has lower wages in the bottom and middle deciles than younger cohorts, but catches up or surpasses the 25–35 cohort in the top decile, which explains the higher Gini coefficient among older workers.

Table 9: Gini Coefficients by Age Group

Age Group	**SILC**	**HBS**
25–35	0.3649	0.3575
35–45	0.4083	0.3648
45–60	0.4641	0.4142

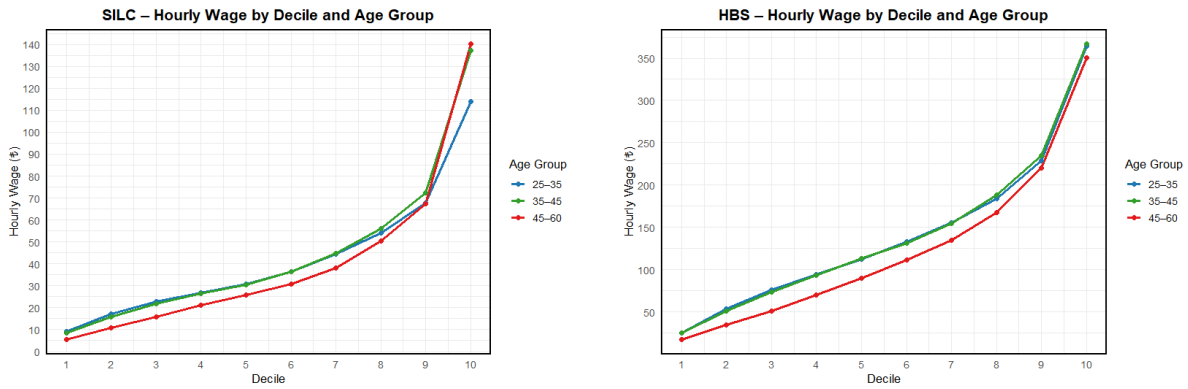


Figure 8: Hourly Wage by Decile and Age Group in SILC (left) and HBS (right)

Furthermore, we analyzed informality in Turkey, which is an important structural factor affecting wage inequality. Using the SILC dataset, summary statistics by employment type were calculated, using registration status with the Social Security Institution (SGK) as the indicator of formal employment (Table 10). The informal employment rate among wage earners in our sample is approximately 22.1%, which has declined to this level over the past two decades according to previous studies in Turkey, such as Bakis and Polat (2021). Notably, informal workers earn less than half as much as formal workers on average (23.59 TL vs. 48.20 TL; median 16.48 TL vs. 36.54 TL).

Table 10: Summary Statistics of Hourly Wages by Employment Type (SILC)

Group	**Mean**	**Median**	**Min**	**Max**	**Share**
Formal	48.20	36.54	1.38	1079.06	77.9%
Informal	23.59	16.48	1.44	807.69	22.1%

Figure 9 illustrates the hourly wage gap between formal and informal workers across deciles in SILC, showing that formal wages exceed informal wages across the entire distribution and diverge sharply at the upper tail. Figure 10 compares the occupation distributions of formal and informal workers: informal workers are heavily concentrated in skilled agricultural, forestry, and fishery work (28.7%), service and sales (22.9%), elementary occupations (17.1%), and craft and related trades (15.6%), whereas formal workers have a much higher share of professionals (20.8

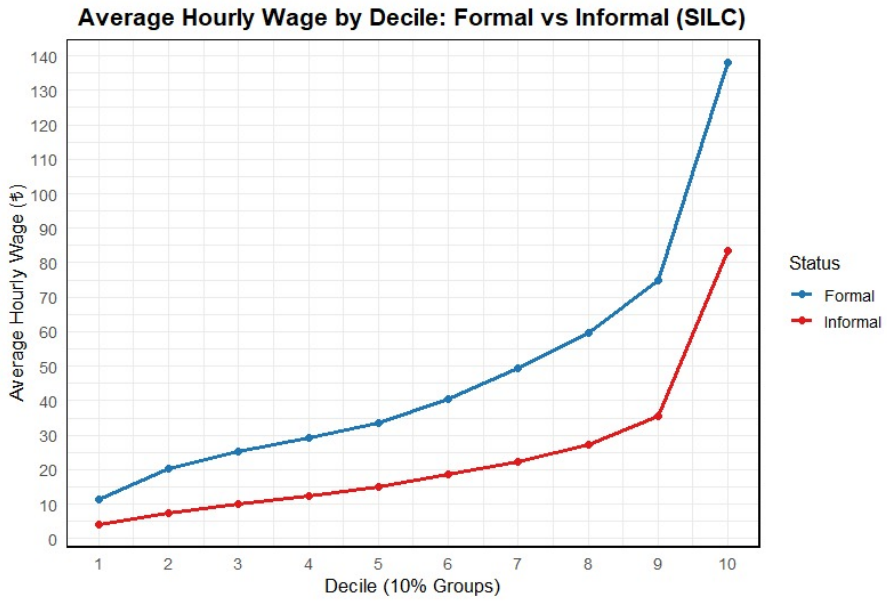


Figure 9: Average Hourly Wage by Decile: Formal vs. Informal Workers (SILC)

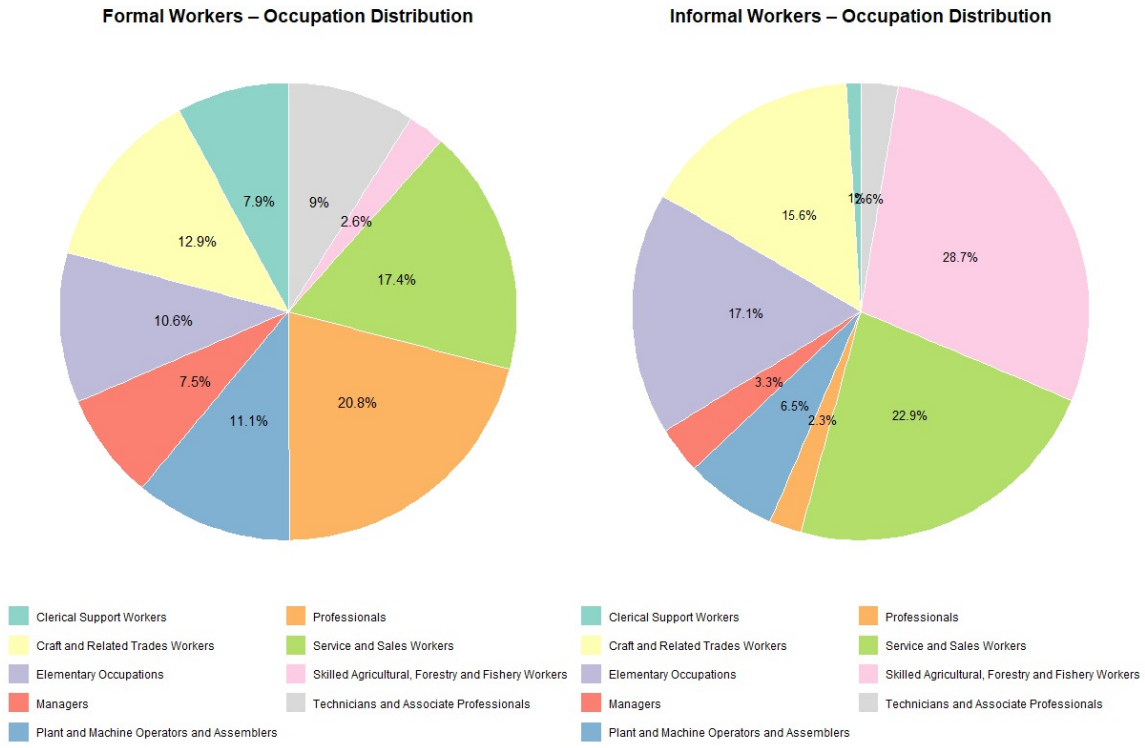


Figure 10: Occupation Distribution of Formal and Informal Workers (SILC)

Finally, we implemented a Mincer-style regression analysis to understand the determinants of log hourly wages. We estimated the relationship between log hourly wage and years of schooling, being female, age, age^2, and regional dummy variables from the SILC

data. Because the HBS dataset does not include regional classification, we estimated the regression without regional controls for HBS:

$$\log(\text{hourly wage}_i) = \beta_0 + \beta_1 \cdot \text{schooling}_i + \beta_2 \cdot \text{female}_i + \beta_3 \cdot \text{age}_i + \beta_4 \cdot \text{age}_i^2 + \sum_{r=1}^{R} \delta_r \cdot \text{Region}_{ir} + \varepsilon_i$$

We used $\log(\text{hourly wage}_i)$ as the dependent variable, where β_0 is the intercept, β_1 is the coefficient for years of schooling, β_2 is the female indicator, β_3 is age, β_4 is age^2, and δ_r captures regional fixed effects. All regions are compared against the reference category, TR1 (Istanbul Region). The estimation results are presented in Table 11.

As expected, years of schooling has a strong and statistically significant positive effect on wages in both datasets (approximately 7.2% higher hourly wages per year of schooling in HBS and 8.6% in SILC), indicating substantial returns to education. Controlling for schooling, age, and region, being female is associated with significantly lower hourly wages (approximately -18.2% in HBS and -18.5% in SILC), showing that while raw average wages of employed men and women are similar, women earn significantly less than men with comparable education and age. The relationship between age and wages follows a classic concave pattern, as shown by the positive coefficient on age (0.1134 in HBS and 0.0806 in SILC) and the negative coefficient on age squared (-0.0013 and -0.0009), suggesting that wages increase with age and experience but at a decreasing rate.

In terms of regional effects, all coefficients for the regional dummies are negative and statistically significant, indicating that workers in other regions earn significantly less than those in Istanbul (TR1). The regional wage gap is especially pronounced in peripheral and less developed regions such as TR9 (East Black Sea, -0.4784), TRB (Central East Anatolia, -0.4226), TRA (Northeast Anatolia, -0.4137), and TRC (Southeast Anatolia, -0.2613). This finding aligns with the observed spatial inequality in Turkey and reinforces the wage disparity between metropolitan and peripheral regions.

	Dependent variable:	
	Log(Hourly Wage)	
	HBS 2022	SILC 2023
	(1)	(2)
Constant	1.8148***	1.2368***
	(0.1522)	(0.0917)
Years of Schooling	0.0716***	0.0856***
	(0.0016)	(0.0011)
Female	−0.1821***	−0.1850***
	(0.0171)	(0.0108)
Age	0.1134***	0.0806***
	(0.0072)	(0.0045)
Age Squared	−0.0013***	−0.0009***
	(0.0001)	(0.0001)
TR2: Batı Marmara		−0.1318***
		(0.0239)
TR3: Ege		−0.2005***
		(0.0193)
TR4: Doğu Marmara		−0.0857***
		(0.0207)
TR5: Batı Anadolu		−0.1009***
		(0.0210)
TR6: Akdeniz		−0.2597***
		(0.0215)
TR7: Orta Anadolu		−0.1630***
		(0.0237)
TR8: Batı Karadeniz		−0.2905***
		(0.0211)
TR9: Doğu Karadeniz		−0.4784***
		(0.0288)
TRA: Kuzeydoğu Anadolu		−0.4137***
		(0.0238)
TRB: Ortadoğu Anadolu		−0.4226***
		(0.0233)
TRC: Güneydoğu Anadolu		−0.2613***
		(0.0207)
Observations	8,979	20,002
R^2	0.2265	0.2913
Adjusted R^2	0.2261	0.2908
Residual Std. Error	0.7108 (df = 8974)	0.6687 (df = 19986)
F Statistic	656.8368*** (df = 4; 8974)	547.6401*** (df = 15; 19986)

*p<0.1; **p<0.05; ***p<0.01. Standard errors in parentheses.

Table 11: Determinants of Log Hourly Wage (Regression Results)

3.2 Comparison with the World

Recent empirical analysis highlights that Turkey exhibits higher levels of wage inequality than most developed countries. To put our 2022 estimates into a global perspective, we compare them against international benchmarks compiled under RED guidelines (Krueger et al., 2010) and recent European evidence (Zwysen, 2024), keeping in mind that all wage premia in Turkey have followed a declining trend over 2002–2022 according to Tamkoç and Torul (2020).

Table 12: Wage Inequality and Wage Premia across Countries

	Level in year 2000				Change				
Country	Var. log w	College premium	Exp. premium	Gender premium	College premium	Exp. premium	Gender premium	Var. log w	Period
Canada	0.45	1.80	1.32	1.33	0.22	0.31	-0.11	0.17	1978-2006
Germany	0.27	1.38	1.27	1.28	-0.08	0.22	-0.15	0.05	1983-2003
Italy	0.17	1.51	1.34	1.03	-0.08	0.11	-0.05	0.03	1987-2006
Mexico	0.62	1.88	1.23	1.21	0.40	0.22	-0.06	0.04	1989-2002
Russia	0.77*	1.50	1.05*	1.49	-0.06	0.05*	-0.07	-0.13*	1998-2005
Spain(a)	0.23	1.48	1.43	1.16	-0.33	0.07	-0.21	-0.18	1985-1996
Sweden(b)	0.18	1.61	1.20	1.22	0.14	-0.02	-0.05	-0.09	1990-2001
UK	0.33	1.62*	1.25*	1.32	0.12*	0.20*	-0.21	0.10	1978-2005
USA	0.44*	1.80*	1.38*	1.36	0.40*	0.28*	-0.25*	0.21*	1980-2006
Average	**0.38**	**1.62**	**1.27**	**1.27**	**0.11**	**0.17**	**-0.10**	**0.04**	

A * indicates the statistic is from data on males only. Wage premia and wage dispersion for women are typically smaller.
(a) Data on changes refer to after-tax annual earnings
(b) Data on levels is for 1992

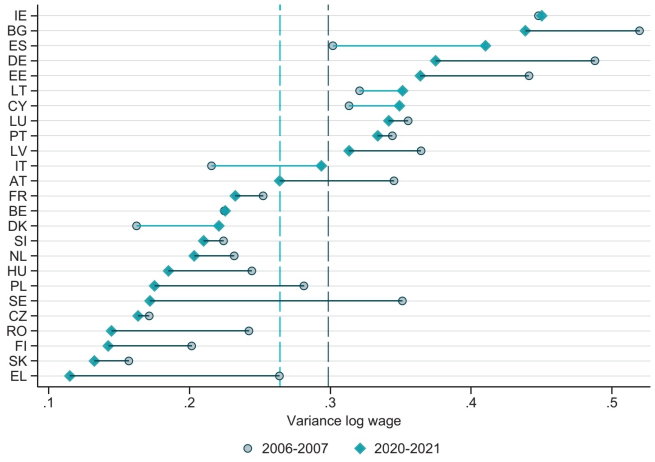
Source: Krueger et al. (2010) using RED guidelines.

Using RED-compliant methods, we found that Turkey's variance of log wages is 0.6304 (SILC) and 0.6983 (HBS), having declined to these levels over the past twenty years (Tamkoç and Torul, 2020). As seen in Table 12, these estimations still exceed the levels observed in most developed countries in the year 2000 (where the cross-country average variance of log wages was 0.38), despite a much lower unadjusted gender wage premium in Turkey (SILC: 1.0461; HBS: 1.0050 vs. an international average of 1.27), which stems from low female labor force participation and selection into formal wage employment (under 27% of wage earners). Thus, even though Turkey has experienced a significant decline in wage inequality over the past two decades, its dispersion remains higher than that of developed economies.

In addition, as Zwysen (2024) notes, Europe has experienced a general decline in wage inequality, similar to Turkey. However, as shown in Figure 11, the variance of log wages across European countries in EU-SILC (ranging roughly from 0.12 to 0.45 in 2020–2021) remains substantially lower than our estimates for Turkey (0.6304 in SILC and 0.6983 in

HBS), even accounting for potential methodological differences.

Figure 11: Change of Variance of Log Wage in Europe (2006–2007 vs. 2020–2021)



Source: Zwysen (2024) via EU-SILC.

Internationally, educational and experience-based wage premiums in Turkey remain important. Our data indicate an education premium of 1.6463 (SILC) and 1.5623 (HBS), and an experience premium of 1.0509 (SILC) and 0.9175 (HBS), highlighting persistent skill-based earnings gaps. Percentile ratios such as P90/P50 (2.5454 in SILC; 2.3535 in HBS) and P50/P10 (2.8448 in SILC; 2.4871 in HBS) further confirm this inequality pattern.

In conclusion, Turkey's wage inequality is not only quantitatively high but also qualitatively distinct. The relatively small unadjusted gender premium masks deeper structural issues, including low female participation, informality, regional disparities, and educational returns. These findings call for policy attention focusing on education quality, formalization of employment, and inclusive labor market reforms.

4 Review of the Literature and Conclusions

In the final section, we review the literature to understand the long-run trend in wage inequality in Turkey and the impact of key macroeconomic and institutional factors: (i) real minimum wage growth, (ii) labor market demand and countercyclical GDP growth, (iii) formalization of employment, and (iv) gender selection in the labor market.

Firstly, it is argued in the literature that real minimum wage growth has a powerful compressing effect on wage inequality, both in Brazil and in Turkey. In the case of

Brazil, Engbom and Moser (2022) show that real minimum wage growth accounts for approximately 45% of the decline in wage inequality while having no significant adverse effect on employment (Figure 12).

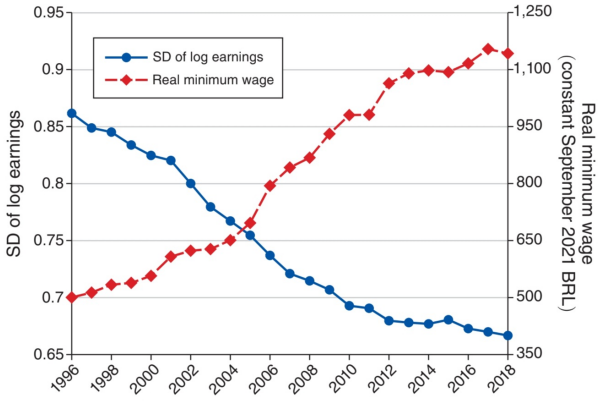


FIGURE 2. EVOLUTION OF WAGE INEQUALITY AND THE REAL MINIMUM WAGE

Notes: Statistics are for males of age 18–54. Real minimum wage is the annual mean of the monthly time series. The correlation between the two time series is −0.973. SD is standard deviation.

Sources: RAIS (1996–2018); IPEA (1996–2018)

Figure 12: Evolution of Wage Inequality and the Real Minimum Wage in Brazil

Source: Engbom and Moser (2022).

This coincides with empirical findings for Turkey demonstrated by Bakis and Polat (2021) and Tamkoç and Torul (2020). They show that the rising real minimum wage is a primary driver of decreasing wage inequality over 2002–2022 because the share of minimum wage earners in Turkey is exceptionally high[2] and average real wages did not increase as rapidly as the real minimum wage. Consequently, the bottom of the wage distribution converged toward the mean, compressing overall wage dispersion. The trajectory of the real net minimum wage from 2002 to 2025 is presented in Table 13. Between 2002 and 2024, the real monthly net minimum wage grew by 238.32%. However, high inflation and the decline in the real minimum wage in early 2025 (from 588.38 TL in 2024 to 549.20 TL in the first four months of 2025 in 2002 real terms) may have reversed some fraction of the reduction in wage inequality achieved between 2002 and 2022.

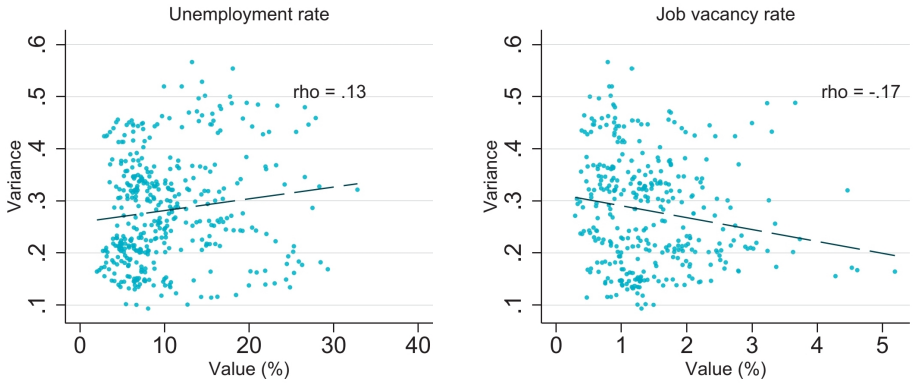
[2]Approximately 40% according to Eurostat.

Year	Annual CPI (2002=100%)	Monthly Minimum Wage Net (Real, 2002 TL)
2002	100.00%	173.91
2003	121.60%	185.85
2004	132.10%	235.17
2005	142.90%	245.03
2006	156.60%	242.95
2007	170.30%	241.39
2008	188.10%	261.78
2009	199.80%	268.67
2010	216.90%	271.02
2011	231.00%	278.98
2012	251.50%	286.47
2013	270.36%	291.59
2014	294.30%	295.11
2015	316.85%	307.66
2016	341.15%	381.35
2017	379.57%	369.91
2018	441.57%	363.05
2019	508.58%	397.36
2020	571.03%	407.11
2021	682.93%	413.79
2022	1176.75%	414.44
2023	1805.59%	551.32
2024	2889.64%	588.38
2025*	4024.76%	549.20

Table 13: Annual CPI and Real Net Minimum Wage in Turkey (2002–2025)

Source: TurkStat. *2025 CPI is calculated via the first 4 months' CPI. The growth rate of the real monthly net minimum wage from 2002 to 2024 is 238.32%.

Secondly, labor market conditions and macroeconomic growth also shape wage dispersion. Zwysen (2024) argues that the unemployment rate and job vacancy rate, which serve as measures of labor demand, exhibit moderate correlations with the variance of log hourly real wages across European country-years ($\rho = 0.13$ for unemployment rate and $\rho = -0.17$ for job vacancy rate, as shown in Figure 13). Stronger labor demand and tighter labor markets tend to compress wage inequality.



Note: correlation between country-year level variance of log hourly real wages and contextual factors at country-year level, adjusted for breaks.
Source: EU-SILC 2007-2022, supplemented with external data.

Figure 13: Correlation between Variance of Log Hourly Real Wages and Labor Market Indicators (Unemployment Rate and Job Vacancy Rate)

Source: Zwysen (2024), EU-SILC 2007–2022.

In addition, GDP growth is negatively correlated with wage inequality in Turkey (i.e., wage inequality is countercyclical), as shown by Tamkoç and Torul (2020) and illustrated in Figure 14. During economic downturns (such as the 2009 crisis), the Gini coefficient spikes, whereas periods of robust GDP growth are associated with declining wage inequality.

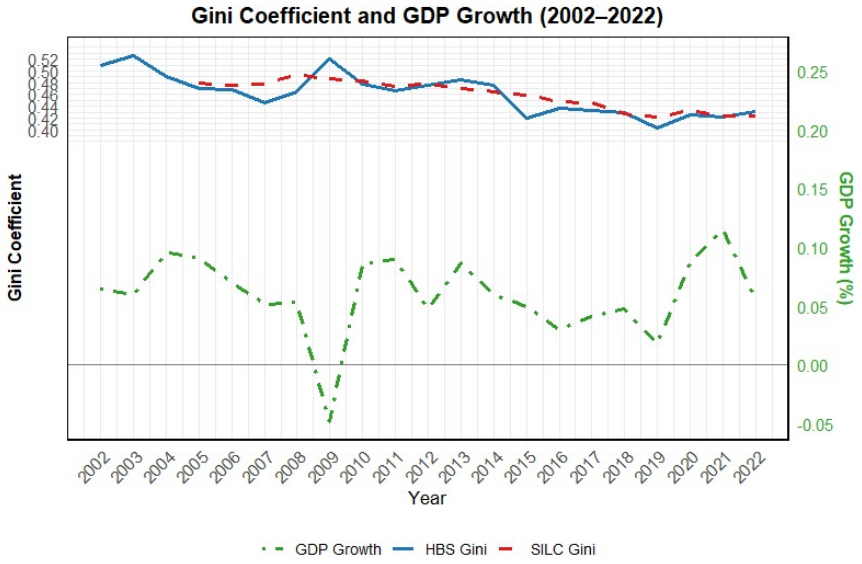


Figure 14: GDP Growth vs. Wage Gini Coefficient Over Time (2002–2022)

Source: World Bank / TurkStat for GDP growth and Tamkoç and Torul (2020, updated version) for Gini coefficients.

Thirdly, the literature highlights the role of declining informality in compressing the wage distribution. As documented by Bakis and Polat (2021), the share of informal workers among positive wage earners in Turkey declined substantially from 29.8% in 2002 to 14.5% in 2019 alongside rising average years of schooling and female share (Figure 15; in our 2022 SILC sample, the informal share is $\sim 22.1\%$ using Social Security Administration registration as a proxy). Tamkoç and Torul (2020) emphasize that most informal workers are located at the bottom of the wage distribution, which is confirmed by our SILC estimation in Figure 16, where 29.7% of informal workers fall in the 1st wage decile and 22.6% in the 2nd decile. Because informal workers earn roughly half the average wage of formal workers, a decline in the informal employment rate mechanically reduces overall wage inequality.

Table 1. Some characteristics of wage earners in Turkey

	Female (%)	Less than secondary share (%)	Years of schooling	Informal share (%)	Firm specific experience for private sector*	Small firm (≤ 10 workers) share for LTS** workers	Large firm (≥ 50 workers) share for ST*** workers
2002	21.1	56.7	8.7	29.8	5.5	47.9	49.7
2003	20.8	54.8	8.9	29.7	5.8	49.2	49.5
2004	21.0	56.9	8.6	31.8	5.5	47.5	47.0
2005	21.3	54.7	8.9	30.6	5.1	45.8	46.8
2006	22.0	53.4	9.0	29.5	4.7	45.2	46.4
2007	22.2	52.4	9.1	27.2	4.7	44.8	45.7
2008	22.8	51.1	9.3	23.8	4.4	42.9	45.6
2009	23.3	50.1	9.4	22.9	4.3	43.9	45.8
2010	23.3	50.8	9.4	22.3	4.1	42.7	45.7
2011	24.0	50.5	9.5	21.9	4.0	43.5	44.6
2012	25.2	48.6	9.6	18.9	4.0	43.8	45.1
2013	26.3	48.1	9.7	17.2	4.0	43.4	45.4
2014	27.0	48.6	9.7	16.2	3.9	45.0	46.7
2015	28.0	47.5	9.9	15.3	3.9	44.8	46.8
2016	28.9	46.0	10.0	14.6	4.1	43.7	47.6
2017	29.4	45.2	10.1	14.7	4.1	44.1	48.5
2018	30.4	44.6	10.1	14.7	4.1	43.0	49.6
2019	31.2	42.7	10.3	14.5	4.4	42.6	50.0

Source: TurkStat, HLFS 2002-19, Only positive wage earners are included in the sample. *Tenure years in current job, ** LTS stands for less than secondary education level, ***ST stands for secondary or tertiary education level

Figure 15: Characteristics of Wage Earners in Turkey (2002–2019)

Source: Bakis and Polat (2021).

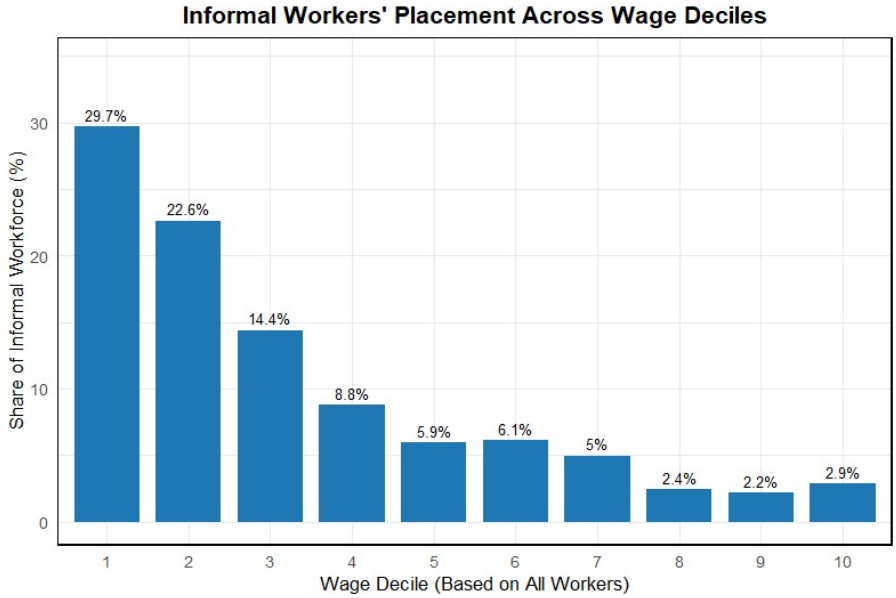


Figure 16: Informal Workers' Placement Across Wage Deciles

Source: Our estimations via SILC (2023).

Finally, it is important to account for the gender factor and sample selection bias. Even though our unadjusted estimation for the gender wage premium in 2022 is approximately 1 (1.0461 in SILC and 1.0050 in HBS)—which at first glance might appear to suggest no wage gap between male and female earners—there is substantial sample selection bias because the sample is restricted to positive wage earners. As shown in Table 14, women account for only 26.9% of wage earners in SILC and 26.4% in HBS. Because women with higher education and skill levels are disproportionately more likely to participate in formal wage employment in Turkey, unadjusted average wage comparisons mask the significant conditional gender wage gap ($\sim 18\%$) revealed in our regression analysis.

Table 14: Share of Female Wage Earners in the Sample

Dataset	**Female Share**
SILC - Female	26.9%
HBS - Female	26.4%

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